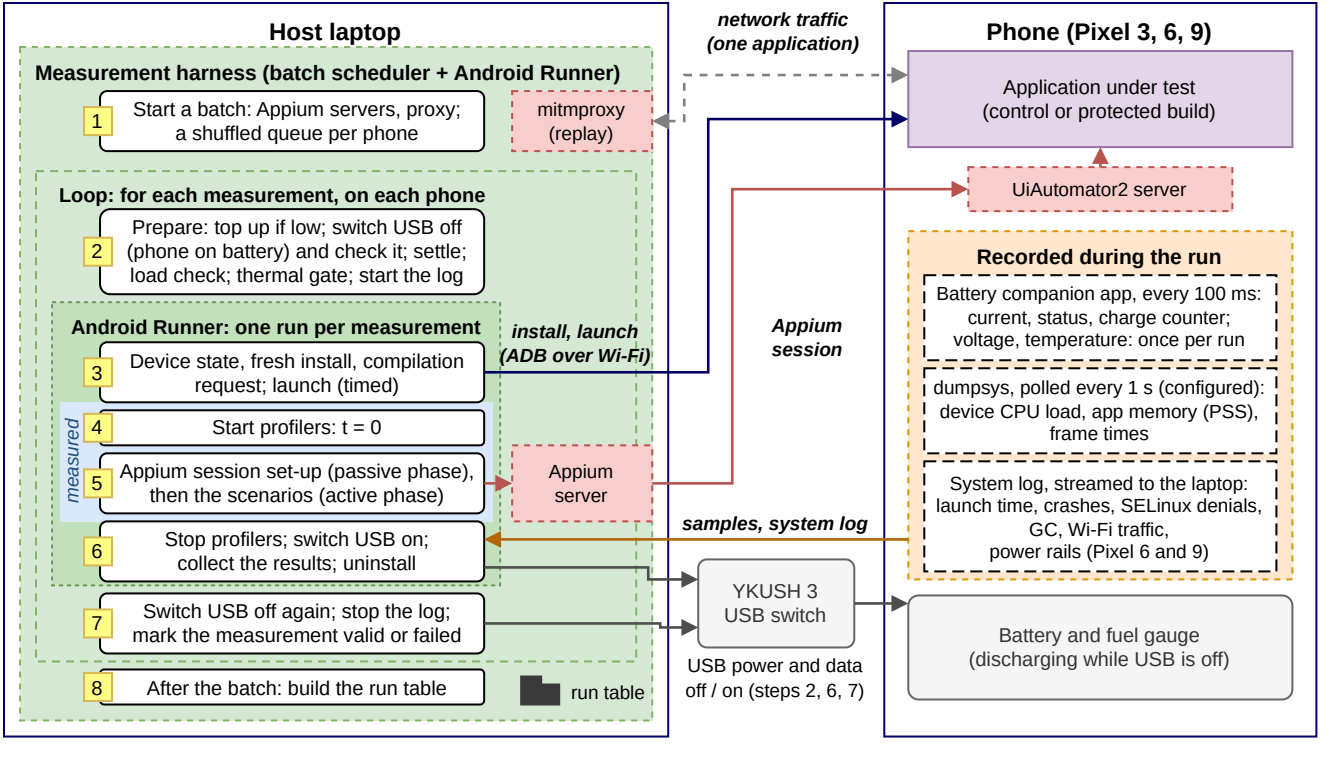

Host laptop
Measurement harness (batch scheduler + Android Runner)
1
Start a batch: Appium servers, proxy; a shuffled queue per phone
mitmproxy (replay)
Loop: for each measurement, on each phone
2
Prepare: top up if low; switch USB off (phone on battery) and check it; settle; load check; thermal gate; start the log
Android Runner: one run per measurement
measured
3
Device state, fresh install, compilation request; launch (timed)
4
Start profilers: t = 0
5
Appium session set-up (passive phase), then the scenarios (active phase)
6
Stop profilers; switch USB on; collect the results; uninstall
7
Switch USB off again; stop the log; mark the measurement valid or failed
8
After the batch: build the run table
run table
network traffic (one application)
install, launch (ADB over Wi-Fi)
Appium session
Appium server
samples, system log
YKUSH 3 USB switch
USB power and data off / on (steps 2, 6, 7)
Phone (Pixel 3, 6, 9)
Application under test (control or protected build)
UiAutomator2 server
Recorded during the run
Battery companion app, every 100 ms: current, status, charge counter; voltage, temperature: once per run
dumpsys, polled every 1 s (configured): device CPU load, app memory (PSS), frame times
System log, streamed to the laptop: launch time, crashes, SELinux denials, GC, Wi-Fi traffic, power rails (Pixel 6 and 9)
Battery and fuel gauge (discharging while USB is off)